

Science & Research

39 words · Science & Research

NOUNS

experiment

a scientific test carried out to discover something or to check whether an idea is true

The scientists repeated the experiment several times to make sure the results were reliable.

discovery

the act of finding something that existed but was previously unknown

The discovery of a new species stunned biologists worldwide.

breakthrough

an important discovery or development that comes after a lot of hard work

Researchers announced a major breakthrough in battery technology this week.

methodology

the set of principles and procedures used to carry out a piece of research

The paper explains the study's methodology in detail so others can repeat it.

sceptic

a person who doubts claims that are generally accepted

She was a sceptic at first, but the data eventually won her over.

theory

an idea or explanation for something that has not yet been proved

It's only a theory at this point. Nobody's proved it yet.

invention

a machine, process, or device created by someone for the first time

The printing press is considered one of the most important inventions in history.

hypothesis

an idea put forward as a possible explanation, to be tested by further study

Her hypothesis was that stress affects memory, and the results supported it.

consensus

general agreement among a group of people, especially experts

There's now a broad scientific consensus on how the disease spreads.

conjecture

an opinion or conclusion based on incomplete information

At this stage it's all conjecture, because nobody actually knows what caused it.

ADJECTIVES

accurate

correct and exact, without errors

The model's predictions have proved remarkably accurate so far.

groundbreaking

introducing new ideas or methods that have a major impact

Her groundbreaking work on vaccines earned her international recognition.

cutting-edge

using the most modern and advanced methods or technology

The university has invested heavily in cutting-edge laboratories.

likely

probable; expected to happen

It's highly likely that we'll see driverless cars on every street within twenty years.

anecdotal

based on personal accounts or individual examples rather than systematic study

The evidence for the remedy is mostly anecdotal, since nobody has tested it properly.

state-of-the-art

incorporating the newest ideas and most up-to-date features

The hospital just installed state-of-the-art scanning equipment.

utopian

relating to an impossibly ideal, perfect society

His vision of a world without work always struck me as hopelessly utopian.

inconclusive

not leading to a definite result or decision

The test results were inconclusive, so they ran the study again.

empirical

based on observation or experiment rather than theory alone

The claim needs empirical support before anyone will take it seriously.

VERBS

prove

to show that something is true using facts or logic

It took the team a decade to prove the theory correct.

predict

to say what you think will happen in the future

Experts predict that the technology will be commonplace within a decade.

debunk

to show that a widely held claim or belief is false

The study debunked the popular myth that we only use ten percent of our brains.

refute

to prove that a statement or theory is wrong

The newly discovered fossils refute the earlier claim that the species died out much sooner.

investigate

to examine a subject closely in order to discover facts

The team is investigating why the reaction happens so much faster at low temperatures.

observe

to watch something carefully in order to gain information

The team observed the same pattern in every trial they ran.

posit

to put forward an idea or theory as a basis for argument

The paper posits that language shaped early human cooperation.

PHRASAL VERBS

find out

to discover a piece of information

It took years of testing to find out what was causing the problem.

come up with

to think of or produce an idea, plan, or solution

It took the team months to come up with a workable solution.

carry out

to perform or complete a piece of work, especially research

The university carried out a five-year study into sleep and memory.

stem from

to originate or develop as a result of something

Many of her ideas stem from years spent observing animals in the wild.

rule out

to decide that something is not a possible explanation or option

The team ruled out contamination before considering other causes.

COLLOCATIONS

trial and error

a method of solving a problem by trying different approaches until one works

The formula was developed through years of trial and error.

a leap forward

a sudden and significant advance or improvement

The findings represent a genuine leap forward in our understanding of the human brain.

take credit for

to accept praise or recognition for something, sometimes without fully deserving it

He was quick to take credit for the discovery, even though his team did most of the work.

IDIOMS

back to the drawing board

used to say that a plan has failed and a new one must be worked out from the start

The experiment didn't work, so it was back to the drawing board.

reinvent the wheel

to waste time and effort creating something that already exists

There's no need to reinvent the wheel when a solution already exists.

get to the bottom of

to find out the real cause or truth of a complicated situation

It took months, but they finally got to the bottom of what was causing the readings to fluctuate.

MODERN SLANG

nerd

a person who is extremely enthusiastic about a particular subject, especially science or technology (informal)

He's a total science nerd who reads research papers for fun.

boffin

a scientist or expert, especially one seen as highly intelligent (informal, British)

The boffins at the lab have finally worked out why it happens.